



MICRO BIRD

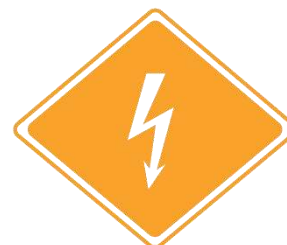
Emergency Response Guide

Micro Bird G5e 140 & 175

Electric School & Commercial Buses

Model Year 2025 to Present

G5e
140 | 175





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Emergency Response Guide



WARNING! ALWAYS wear full Personal Protection Equipment (PPE) including a Self-Contained Breathing Apparatus (SCBA) when fire is involved. Fires in crash-damaged Electric Vehicles (EV) could emit toxic or combustible gases. Small amounts of eye, skin, or lung irritants may be present. If exposed, rinse with large amounts of water for 10–15 minutes. Always consider the entire bus as energized. PPE should include: Class 0 electrical insulating gloves, safety goggles, and Electrical Hazard (EH) rated safety shoes or boots, which must be at least appropriate up to 1000 volts AND a non-conductive object approximately 5 ft (1.5 m) long to safely push someone away from the bus if they accidentally come in contact. Failure to comply can result in serious injury or death.

WARNING! After deactivation, the High Voltage (HV) propulsion system requires 10 minutes to de-energize. Failure to acknowledge can result in serious injury or death.

WARNING! Regardless of the disabling procedure you use, ALWAYS ASSUME THAT ALL HIGH VOLTAGE COMPONENTS ARE ENERGIZED. Cutting, crushing, or touching high voltage components can result in serious injury or death.

WARNING! Do NOT handle a submerged bus without the appropriate PPE for water rescue. Failure to comply can result in serious injury or death.

WARNING! There may be a short circuit if High Voltage (HV) equipment or High Voltage (HV) cables (orange sheathing) are damaged due to an accident related to the equipment. Put on insulated PPE such as insulated clothes and gloves before beginning rescue operations. Failure to comply can result in serious injury or death.

WARNING! The bus should be lifted or manipulated only if first responders are trained and equipped at the technician level per the applicable country national fire training requirements and are familiar with this bus's lifting points. Never come into contact with the High Voltage battery or other High Voltage (HV) components. Failure to comply can result in serious injury or death.

WARNING! Never attempt to lift the bus by the tow hooks. Tow hooks are designed for horizontal pulling only, NOT FOR LIFTING. Failure to comply can result in serious injury or death.



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WARNING! Though the High Voltage (HV) systems are designed to be automatically disabled when the Master Service Disconnects (MSD) are removed, it is strongly recommended to only remove the Master Service Disconnects (MSD) AFTER the ignition key is removed and the 12V battery disconnect switch is turned to the OFF position. This further reduces the chance of an ARC FLASH EVENT which is possible while there is a load on the circuit. Failure to comply can result in serious injury or death.

WARNING! Flames, smoke, arcing, or hot spots like melted plastic may indicate fire or the presence of High Voltage (HV). A thermal camera or Infrared (IR) temperature probe may be useful to identify hot spots. Clear the area around the bus (when possible) and open the bus doors to avoid build-up of gases. If a battery has begun burning, it will try to burn to completion. Cool down the batteries with direct and large amounts of water. If possible, remove combustible materials from around the bus to prevent fire growth. Failure to comply can result in serious injury or death.

WARNING! Take appropriate measures to protect civilians downwind from an incident. Civilians and passengers should be kept at least 50 ft (15 m). away from the bus after evacuation. Failure to comply can result in serious injury or death.

WARNING! Do NOT use foam to extinguish a fire on electric buses. Failure to comply can result in serious injury or death.

WARNING! Use large amounts of water to cool the High Voltage (HV) battery packs if they catch fire, are exposed to high heat, or are generating heat or gases,. If water is not immediately available, use CO₂, dry chemicals, or another typical fire-extinguishing agent to fight the fire until water is available. Failure to comply can result in serious injury or death.

WARNING! If the bus is submerged, do NOT remove the Master Service Disconnects (MSD). Failure to comply can result in serious injury or death.

WARNING! The parking pawl becomes ineffective when the driveshaft is removed. Chock the wheels under this condition. Failure to comply can result in serious injury or death.

WARNING! Do NOT secure the driveshaft to any High Voltage (HV) or Low Voltage (LV) wiring, cooling hoses, or brake lines. Failure to comply can result in serious injury or death.



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WARNING! Damaged buses should be isolated outdoors until inspected. Separate the bus from all combustibles and structures by a distance of at least 50 ft (15 m). Failure to comply can result in serious injury or death.

WARNING! Lack of engine noise does not mean vehicle is off. Silent movement or instant restart capability exists until vehicle is fully shut down.

CAUTION! Moving the bus with the rear wheels on the ground and driveshaft not removed, even for a short distance, can cause the drivetrain motor to overheat and lead to irreversible damage of the motor and possibly start a fire.



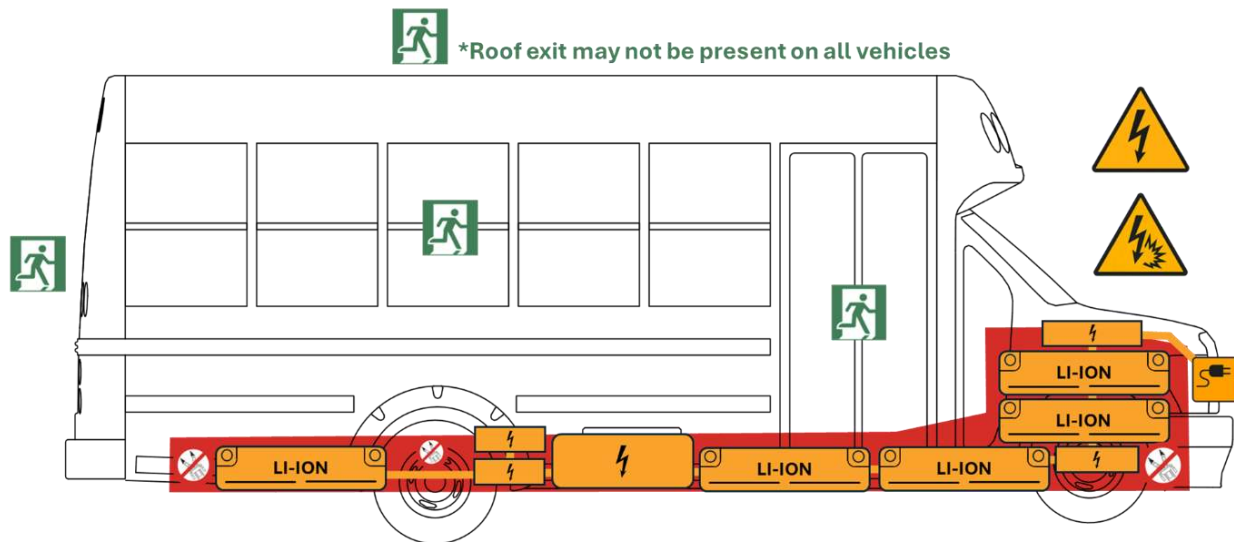
Occupational Health and Safety Administration (OSHA) refers to an Arc Flash as a phenomenon where a flashover of electric current leaves its intended path and travels through the air from one conductor to another or to the ground. The results are often violent and when a human is in close proximity to the Arc Flash, serious injury and death can occur.



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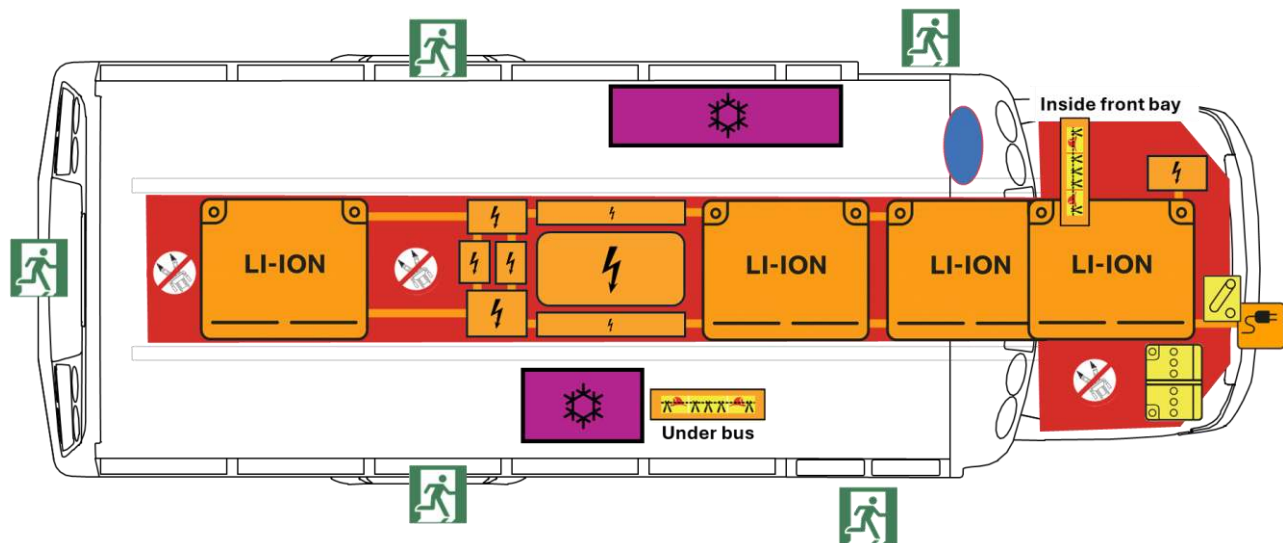


0. Rescue / Emergency Response Sheet



Do NOT cut into or penetrate the region highlighted in red as High Voltage (HV) can be present at any time

NEVER CUT HIGH VOLTAGE CABLES (ORANGE SHEATHING)



WARNING! There may be a short circuit if High Voltage (HV) equipment or High Voltage (HV) cables (orange sheathing) are damaged due to an accident related to the equipment. Put on insulated PPE such as insulated clothes and gloves before beginning rescue operations. Failure to comply can result in serious injury or death.



1. Identification / Recognition

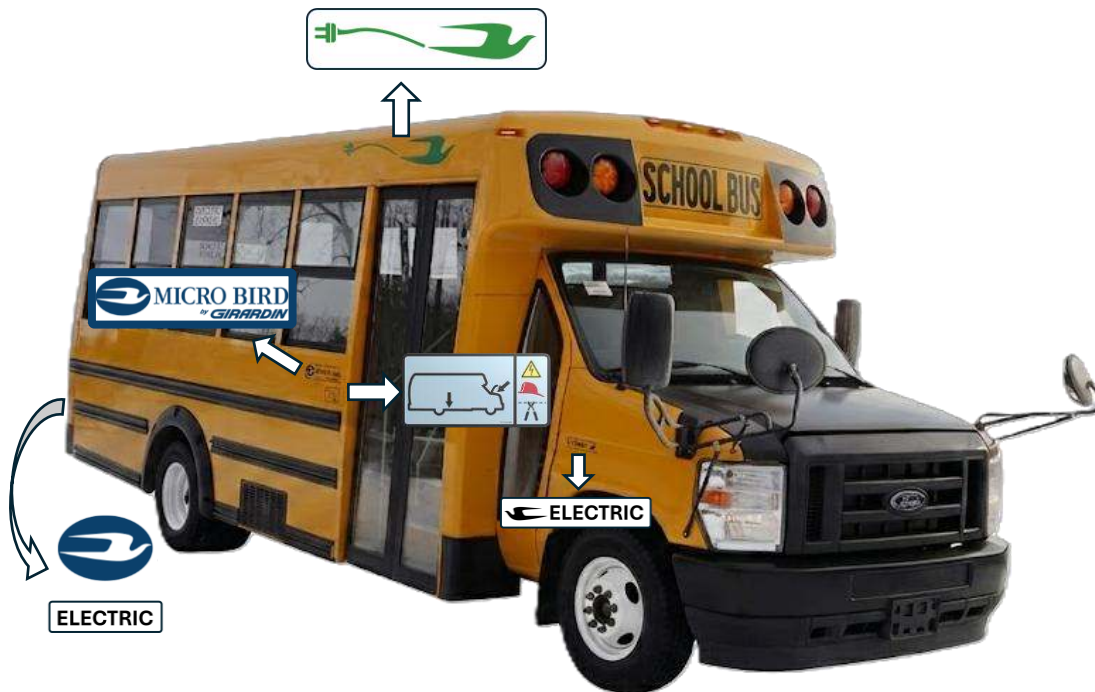
Exterior

Micro Bird G5e buses are built on Ford E-450 cutaway chassis. They can be identified based on a few specific visual clues on the side and rear faces of the vehicle.

This vehicle is offered in two configurations:

- G5e 140 kWh has a high-voltage battery system made of 4 LFP packs with a max voltage of 700 V
- G5e 175 kWh has a high-voltage battery system made of 5 LFP packs with a max voltage of 660 V

Micro Bird electric buses are identified by a decal located on the passenger side of the vehicle near the double-opening door. It also has a flying bird in a blue oval logo on the passenger side at the rear of the vehicle.



A green or blue flying bird followed by an electric cord large logo may appear near the roof at the front on both sides of the bus.

A black square decal showing a flying bird and the text ELECTRIC can be found on both sides of the vehicle over the arches of the front wheels.

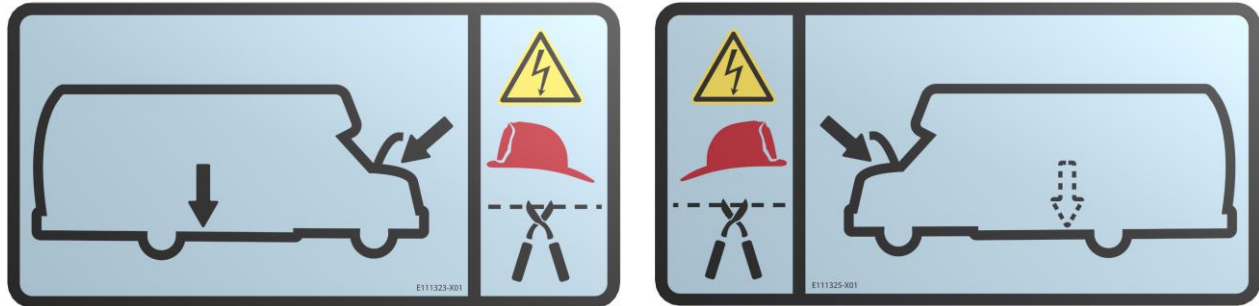
A sticker on each side of the bus, near the service door and the driver's door, indicate where the first responder loops are located. One way to differentiate the older generation of electric vehicles (G5e



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88 kWh) from the current (G5e 140/175 kWh) is from the locations of the first responder loops. Older generation (G5e 88 kWh) has only one loop located in the front bay of the vehicle. G5e 140/175 kWh buses have two (2) first responder loops: one in the front bay and one on the passenger side under the bus, close to the electric motor.



Both G5e 140 & 175 buses have a radiator grille located on each side, close to the rear wheels.





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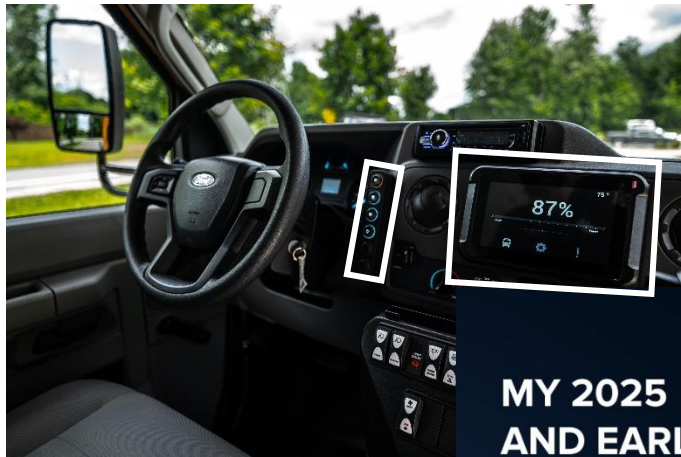


Interior

All Micro Bird electric buses are equipped with a touch screen located on the dashboard above the console.

Micro Bird G5e 140 & 175 electric buses have a PRND controller with six (6) illuminated buttons located on the right side of the steering column.

Some early G5e electric buses also have a left-hand gauge in the cluster assembly having the words CHARGE – ECO – POWER marked on it, along with a flying bird logo printed under the ECO marking.



**MY 2025
AND EARLIER**



**MY 2026
AND LATER**





2. Immobilization / Stabilisation / Lifting

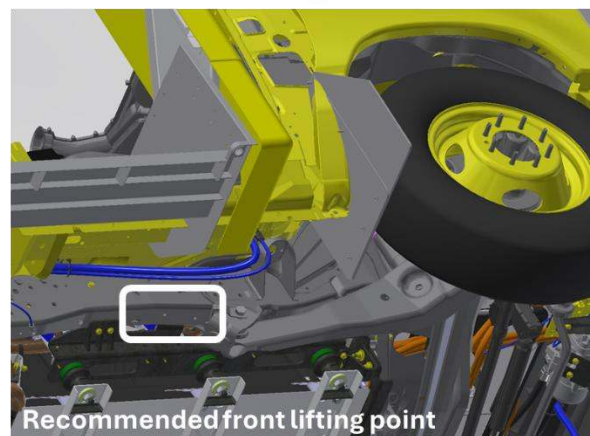
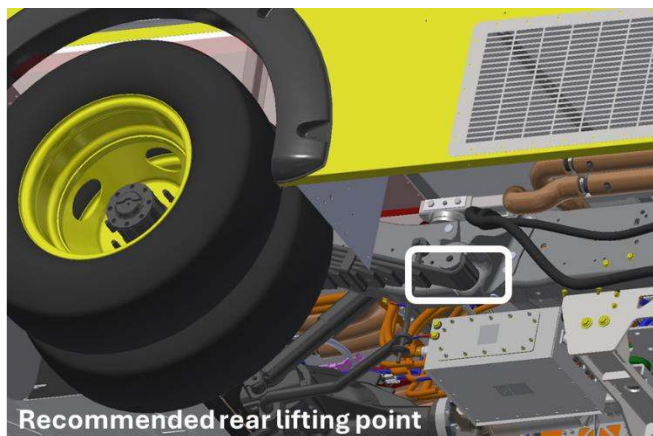
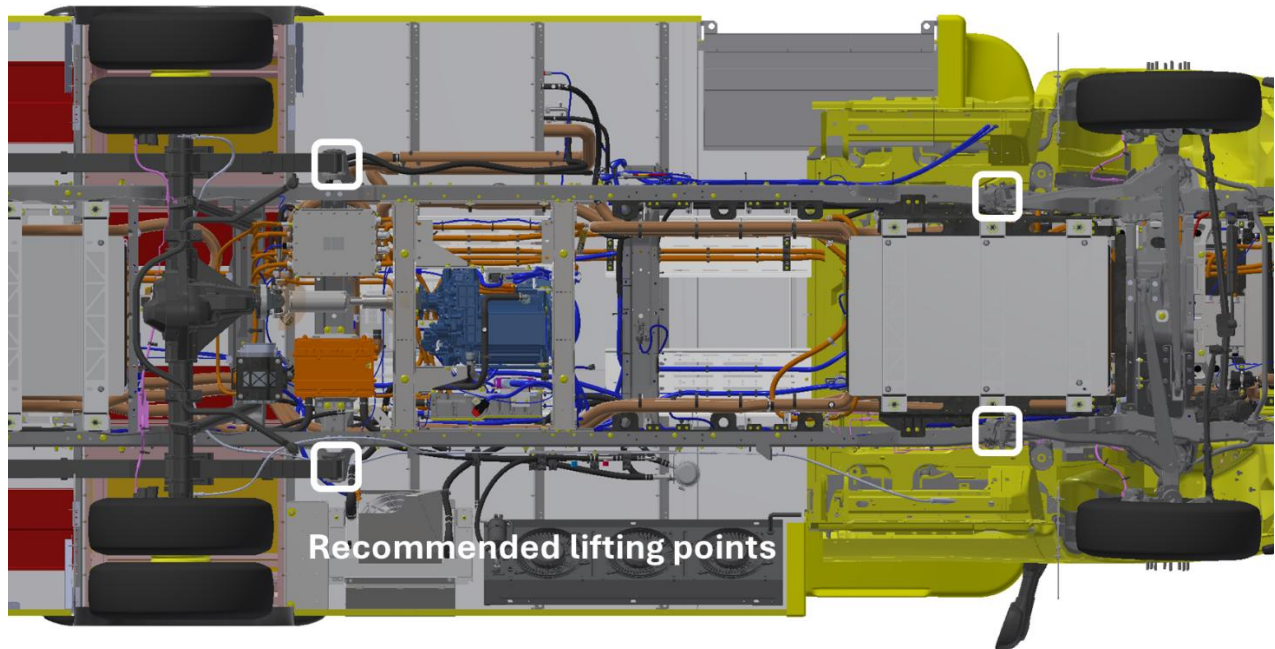
Immobilization / Stabilization

Step 1 - Firmly press the parking brake pedal (located to the left of service brake pedal) to the floor

Step 2 - Chock the wheels

Lifting

Micro Bird buses should only be lifted at the axles using jacks of sufficient capacity. It is recommended to place the jacks securely at the suspension joints. It is also strongly recommended to additionally support the bus with blocks or jack stands under the frame rails when lifted. One of these additional supports should be located under one of the chassis rails behind the rear axle of the vehicle.

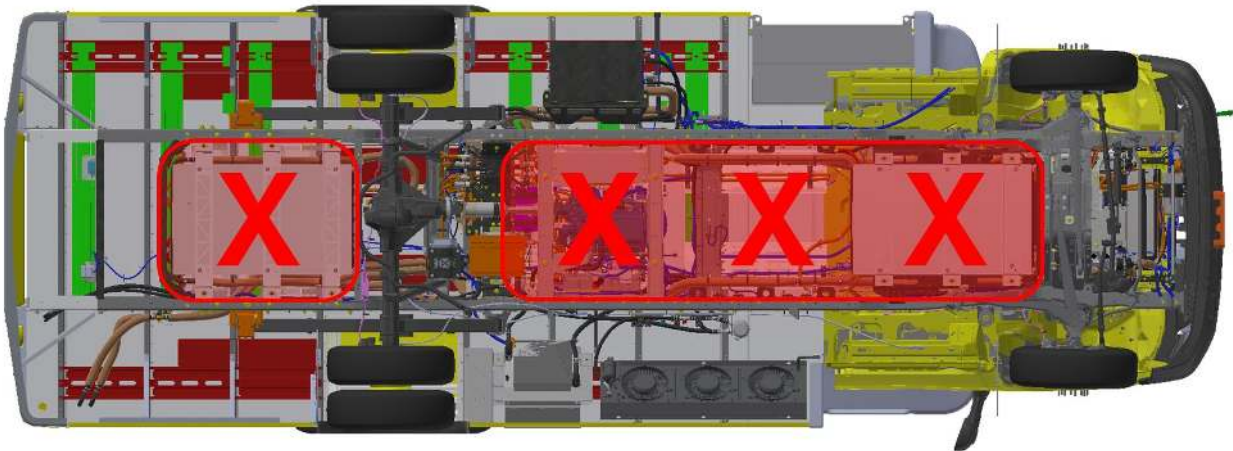




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Never position jacks or blocks under the battery packs or the motor support to lift the vehicle as this may lead to damages to the powertrain along with potentially serious injury or death.



WARNING! The bus should be lifted or manipulated only if first responders are trained and equipped at the technician level per the applicable country's national fire training requirements and are familiar with this bus lifting points. Never come into contact with the High Voltage (HV) battery or other High Voltage (HV) components. Failure to comply can result in serious injury or death.

WARNING! Proper jacking procedures and basic safety measures must be observed to ensure safety of personnel while working under the bus. Always check the serviceability of any lifting equipment prior to use. Ensure that the lifting device is of sufficient strength to handle the bus and that the surface provides the necessary firmness to support the weight of the bus concentrated on the footprint of the jack. Never move under a bus supported only by a hydraulic jack. Failure to comply can result in serious injury or death.

WARNING! Never attempt to lift the bus by the tow hooks. Tow hooks are designed for horizontal pulling only, NOT FOR LIFTING. Failure to comply can result in serious injury or death.

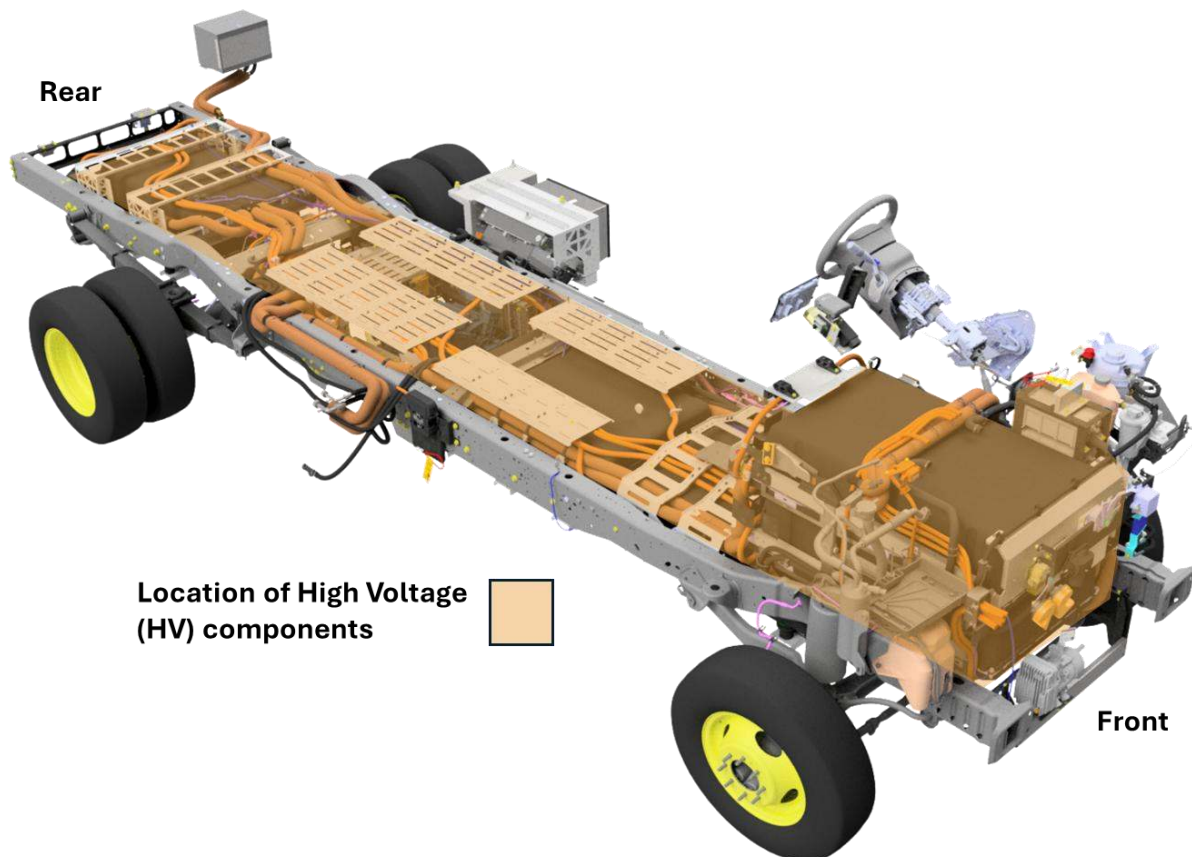


3. Disable Direct Hazards / Safety Regulations

The HV system is electrically isolated from the chassis and monitored by an isolation monitoring device. This further prevents the danger of electric shock should one of the High Voltage (HV) wires be severed and brush up against the bus. This safety mechanism is in addition to the High Voltage Interlock Loop (HVIL) system that, upon detection a severed cable, shuts the system down.

WARNING! Shutting down the High Voltage (HV) system will not shut down the Low Voltage (LV) system as a charged 12V battery will continue to provide power to this system. It is **MANDATORY** to shut down both the High Voltage (HV) and Low Voltage (LV) systems before performing maintenance on the electrical system of the vehicle. Failure to comply can result in serious injury or death.

WARNING! Shutting down the Low Voltage (LV) system will not automatically shut down the High Voltage (HV) system as the DCDC converters will continue to provide power to the Low Voltage (LV) system as long as the High Voltage (HV) battery is charged. It is **MANDATORY** to shut down both the High Voltage (HV) and Low Voltage (LV) systems before performing maintenance on the electrical system of the vehicle. Failure to comply can result in serious injury or death.





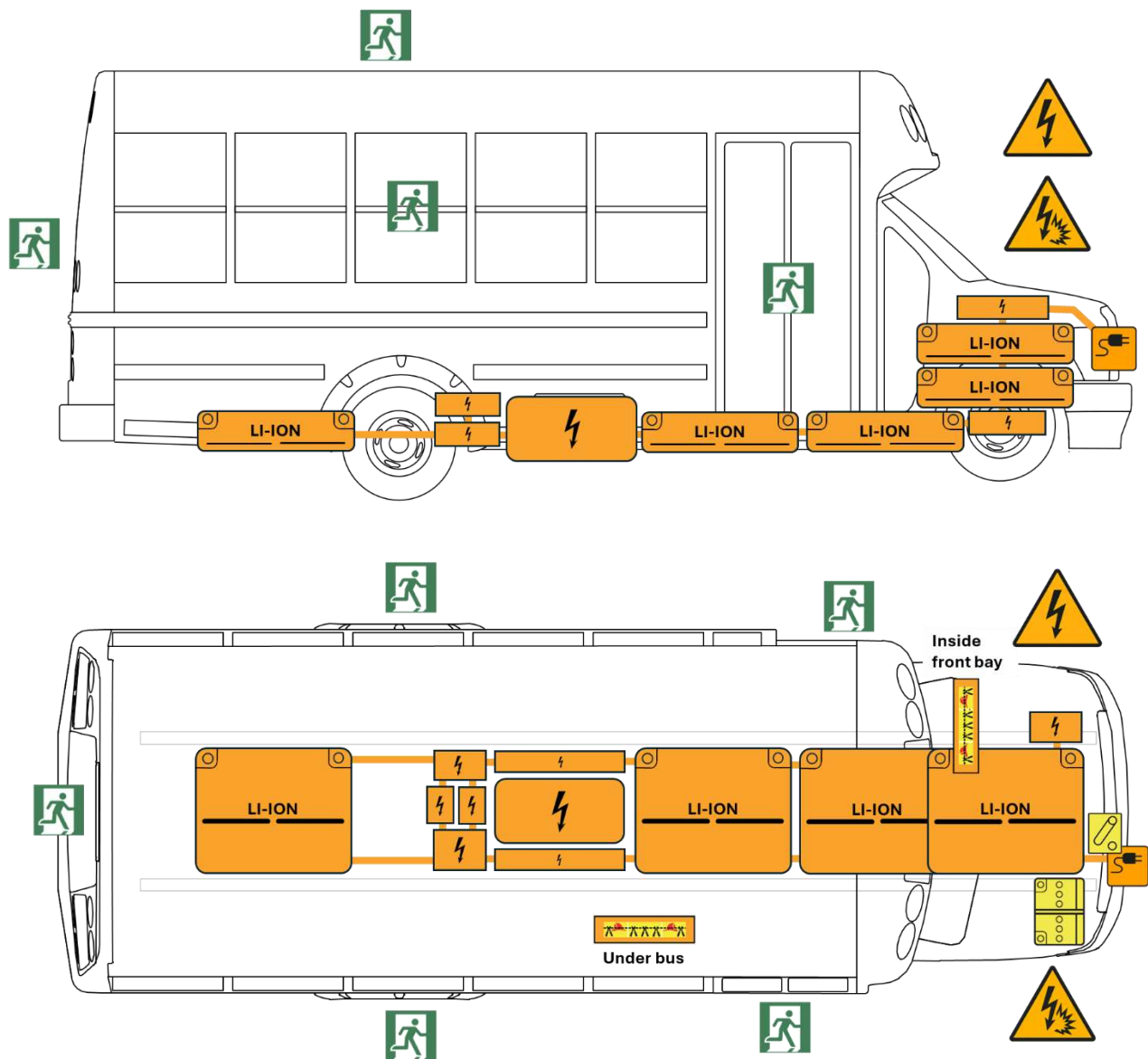
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All High Voltage (HV) components and wires are located within the limits of the rails of the chassis and in the front bay of the vehicle. G5e 140 & 175 buses have two (2) high-voltage sub-systems:

- The battery system made of four (4) or five (5) battery packs along with a Switch Box (S-Box). This part of the High Voltage system is always active unless the Master Service Disconnects (MSD) are removed from the Switch Box (S-Box) and each individual pack
- The main high-voltage system providing power to the propulsion, heating, HVAC and low-voltage system. This part of the system can be disabled from the touch screen located close to the driver or by disconnecting/cutting at least one of the First Responder Loops.

These systems are mixed within the internal area bordered by the longitudinal rails of the chassis and the front bay of the vehicle.





WARNING! Never attempt to access the High Voltage (HV) battery system connections. Never cut or alter a wire of the battery system. This can only be done under the direct supervision of a Micro Bird or Ecotuned representative. Failure to comply can result in serious injury or death.

Removing cable from the charge inlet

Method 1A – AC charge

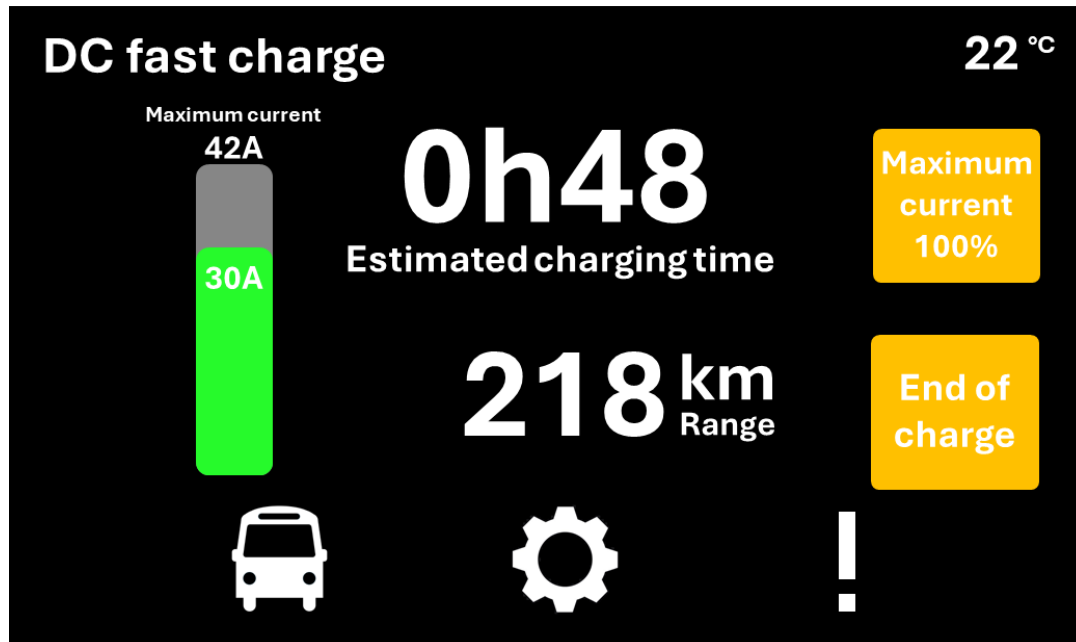
- Step 1 - Press the button on the charge connector
- Step 2 - Remove connector from the charge inlet

Method 1B – DC charge

- Step 1 – Start by shutting down the charging station if it's operating in fast charging mode.
- Step 2 - Press button on the charge connector
- Step 3 - Remove connector from the charge inlet

Method 2 – if previous method failed for DC charge

- Step 1 – Verify ignition is ON and screen is active.
- Step 2 – Select the end of charge button on the touch screen near to the driver seat



- Step 2 - Remove connector from the charge inlet



Method 3 – if previous methods failed for DC charge

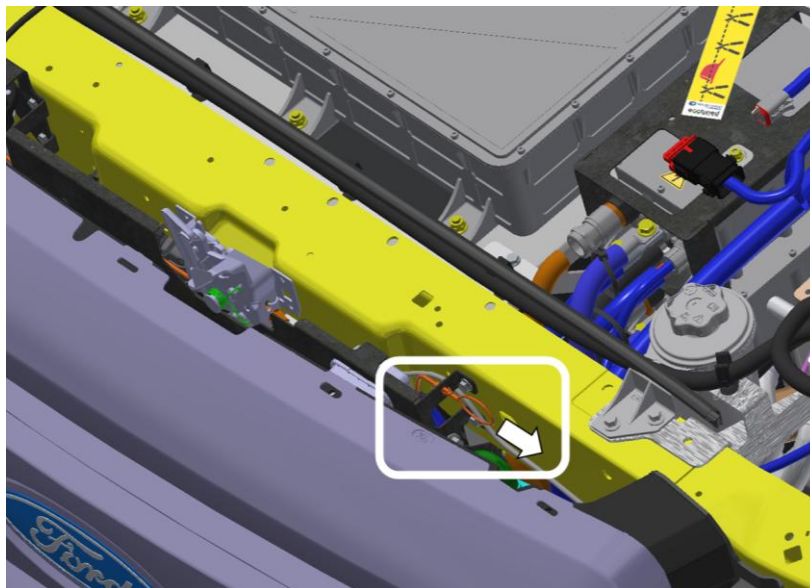
Make sure the charge is ended

Step 1 - Open the front bay

Step 2 – Remove the plastic trim behind the front grille



Step 3 - Locate the pull cable handle on the driver side near the front grille



Step 4 - Remove connector from the charge inlet while pulling handle



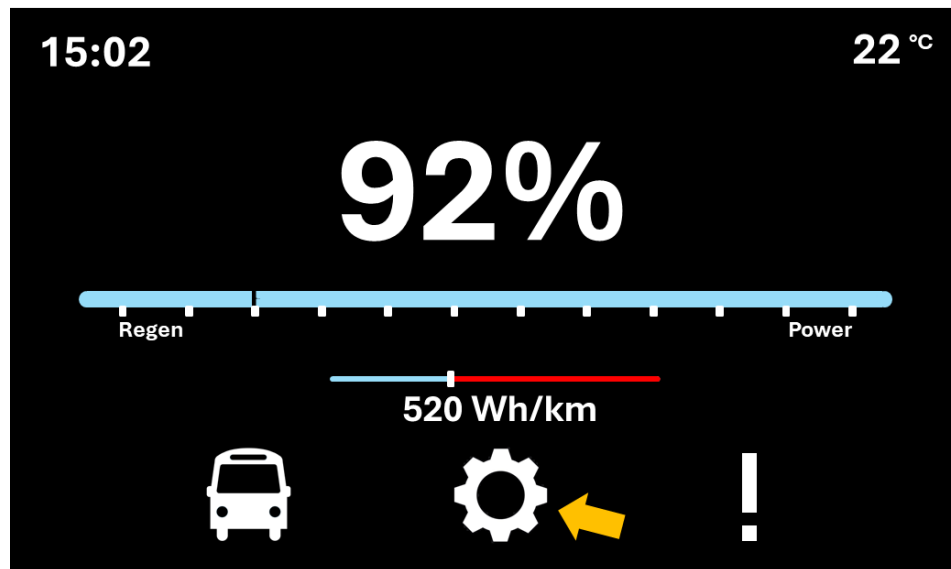
High Voltage (HV) electrical system shutdown

A flow chart at the end of this section is provided for quick reference.

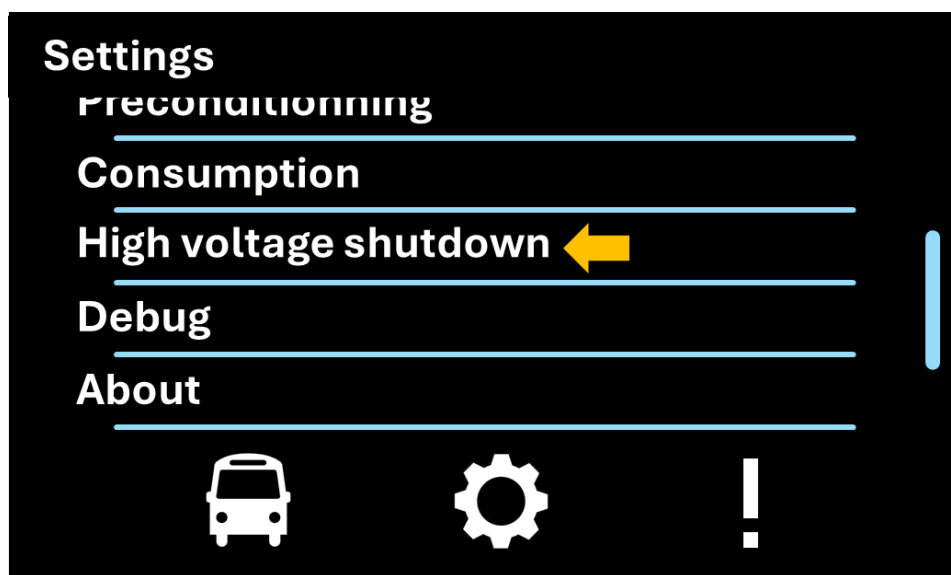
Method 1- if vehicle interior is easily and safely accessible

Step 1 - Verify ignition is ON and the screen is active

Step 2 - Select the gear in the lower portion of the screen



Step 3 - Select the **High voltage shutdown** menu item

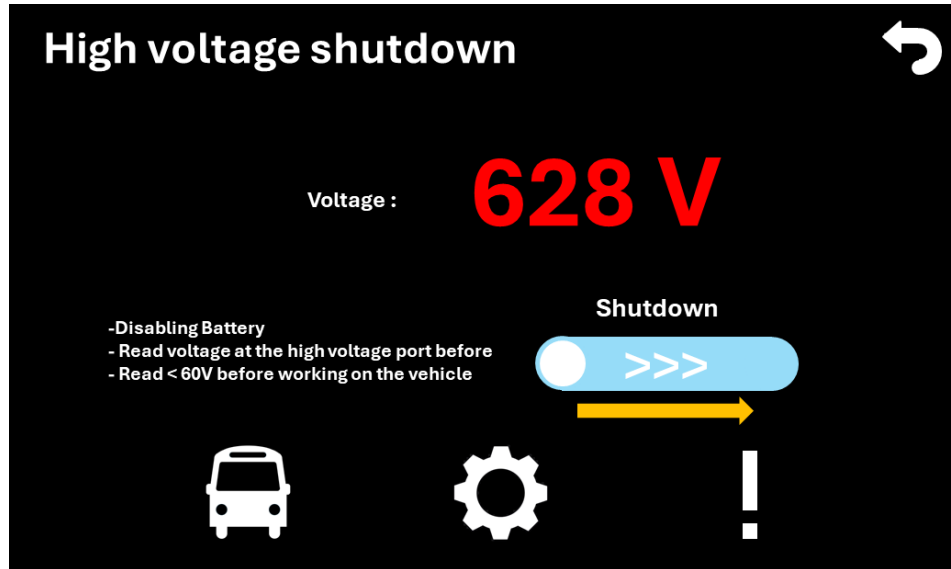




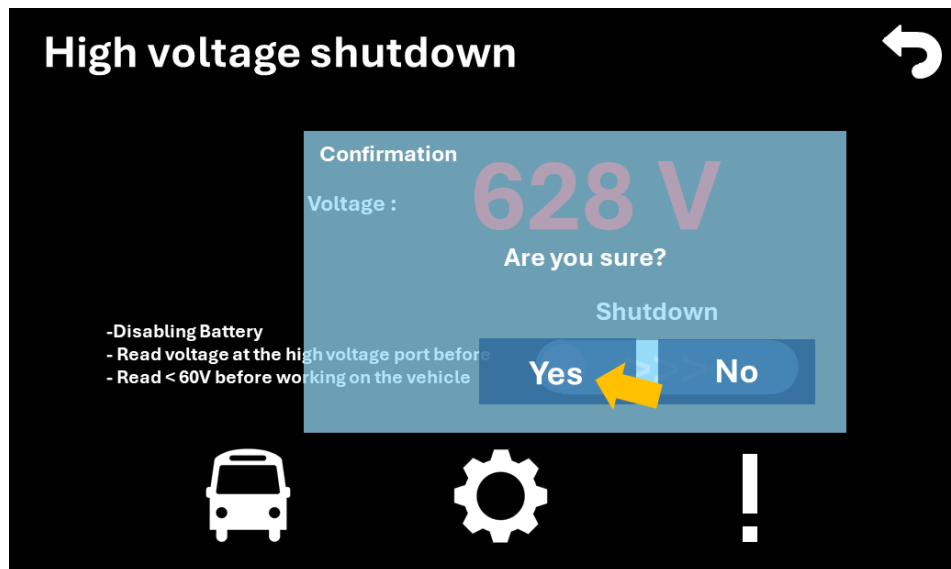
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Step 4 - Slide the shutdown control to the right to access the confirmation panel



Step 5 - Confirm High Voltage (HV) shutdown

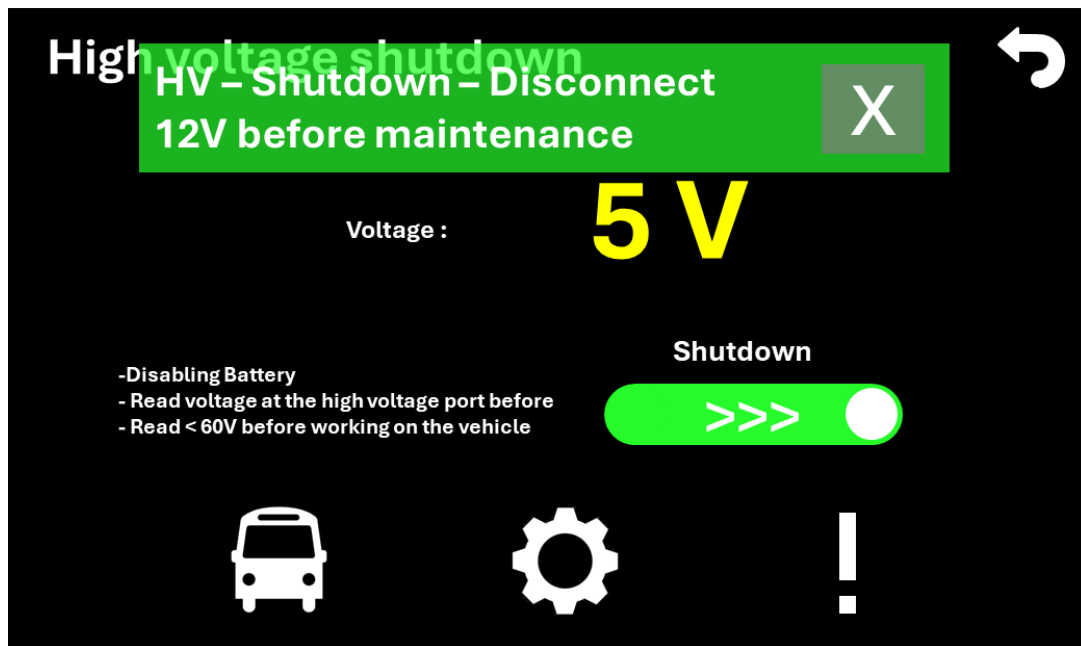




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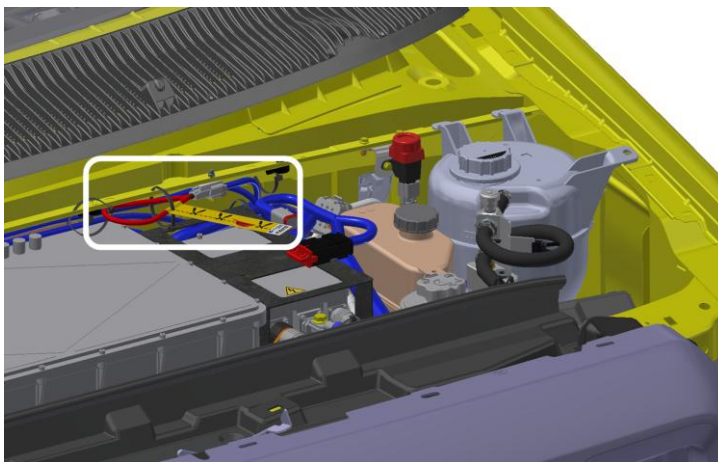


Step 6 – Notice the display voltage dropping. Wait for at least 10 minutes for residual High Voltage (HV) power to dissipate. Residual voltage may be present in capacitors for up to 10 minutes after shutdown.



Method 2 – if vehicle interior is not easily accessible but front bay can be opened

Step 1 - Locate the First Responder Loop in the front bay of the vehicle



Step 2 - Either cut the loop or pull it out of its socket

Step 3 - Wait for at least 10 minutes for residual high-voltage power to dissipate. Residual voltage may be present in capacitors for up to 10 minutes after shutdown.

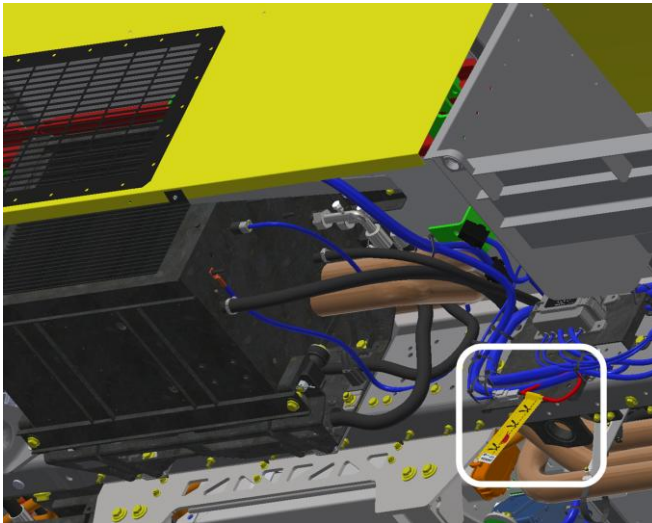


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Method 3 – if vehicle interior is not easily accessible and front bay cannot be opened

Step 1 - Locate the First Responder Loop under the side skin on the passenger side of the bus



Step 2 - Either cut the loop or pull it out of its socket

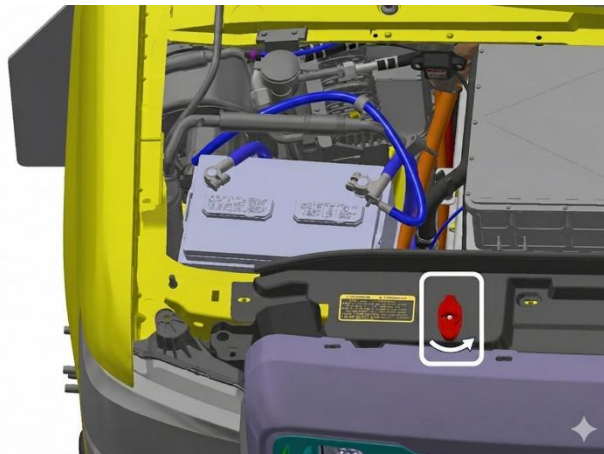
Step 3 - Wait for at least 10 minutes for residual high-voltage power to dissipate. Residual voltage may be present in capacitors for up to 10 minutes after shutdown.

Low-voltage electrical system shutdown

Low-voltage system can only be deactivated in the front bay of the vehicle.

Method 1- if vehicle is equipped with a cutoff switch

1. Turn OFF the ignition key and remove the key from barrel
2. Turn OFF the cutoff switch (turn counterclockwise) in the front bay of the vehicle

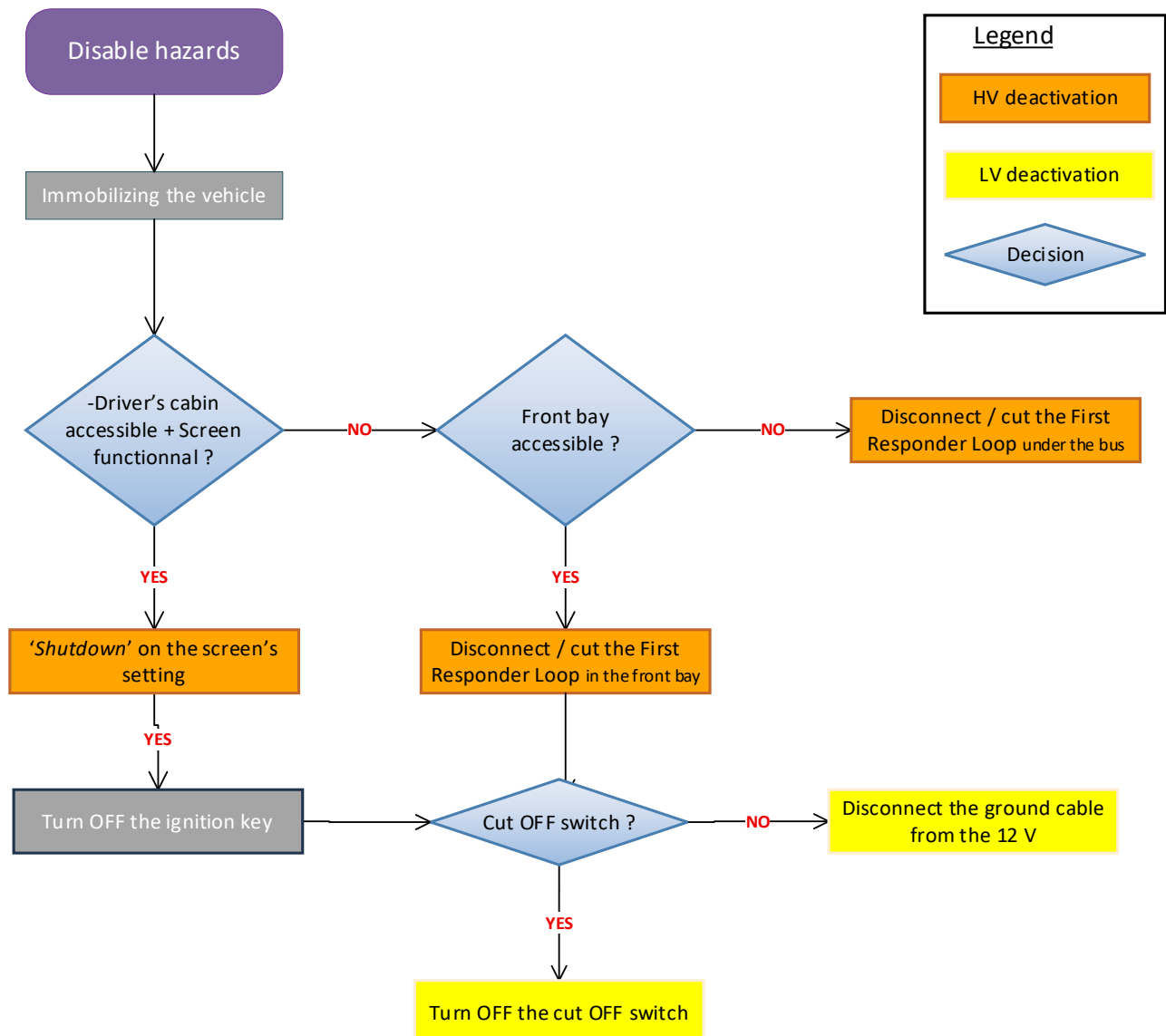




Method 2- if vehicle is not equipped with a cutoff switch

1. Turn OFF the ignition key and remove the key from barrel
2. Using a 13 mm wrench disconnect the ground cable from the battery in the front bay of the vehicle

Flow chart of disable direct hazards



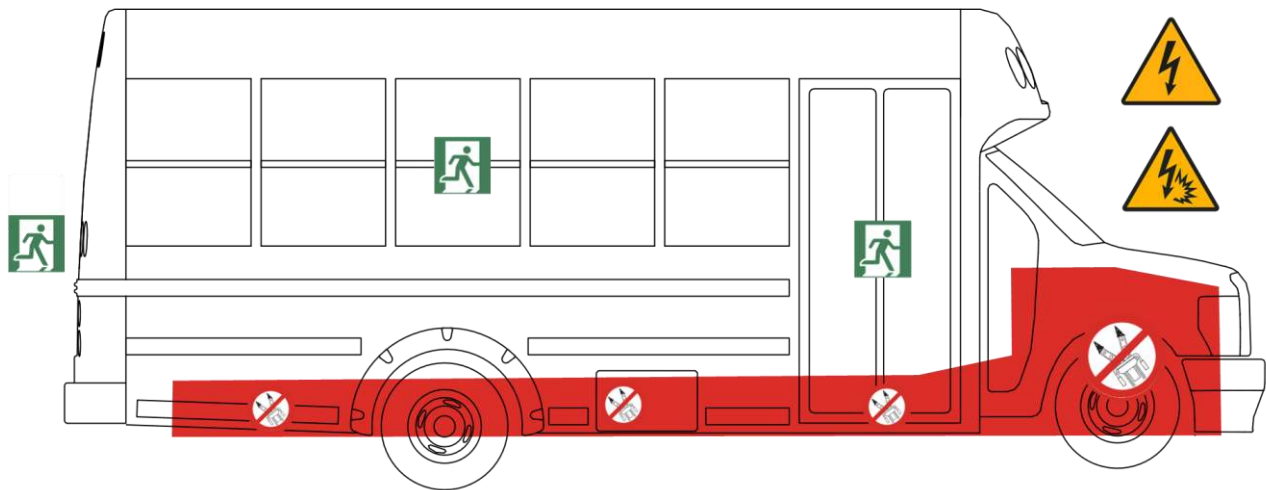


4. Access to the Occupants

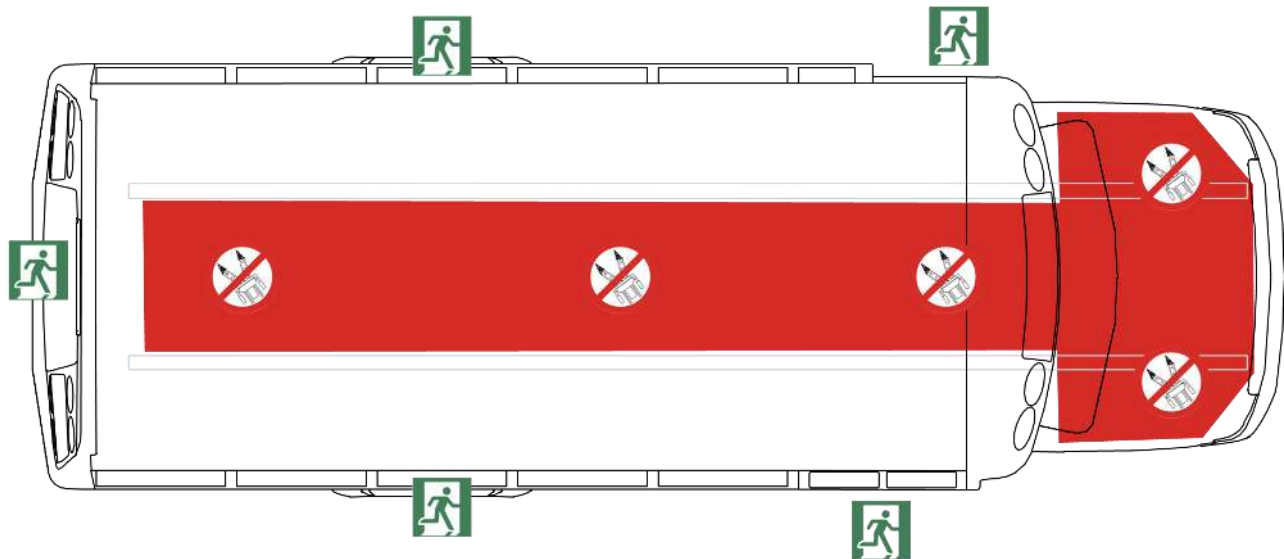
WARNING! The battery High Voltage (HV) system may not be deactivated after an accident, even after the opening / removal of the first responder loops. Do not cut any orange sheathed wire as failure to comply can result in serious injury or death.



*Roof exit may not be present on all vehicles



Do NOT cut into or penetrate the region highlighted in red as High Voltage (HV) can be present at any time



NEVER CUT HIGH VOLTAGE CABLES (ORANGE SHEATHING)



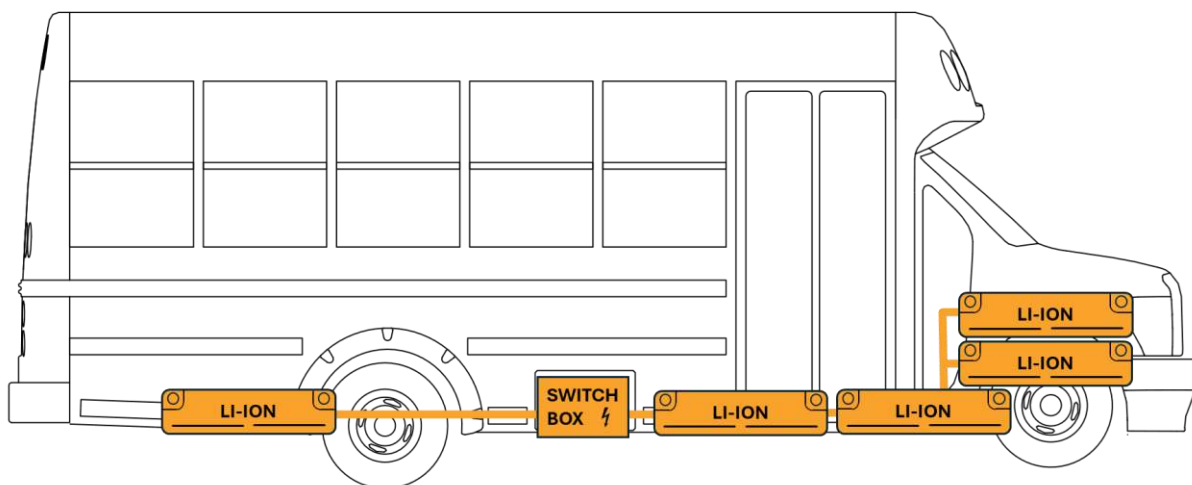
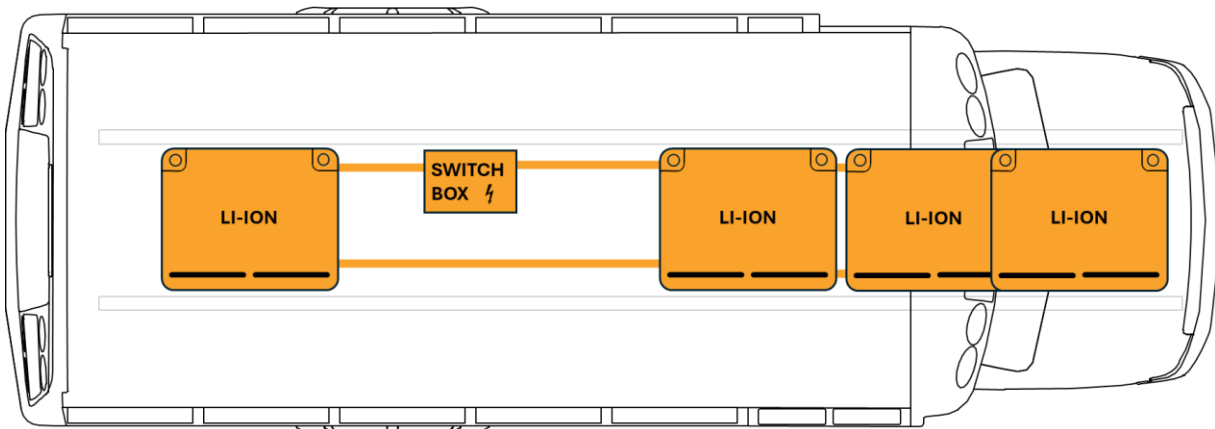
5. Stored Energy / Liquids / Gases / Solids

High Voltage (HV) components are identified by High Voltage (HV) symbols, decals and / or blaze-orange cabling.

G5e 175 kWh

The High Voltage battery system is composed of five (5) lithium-ion packs built from Lithium-Iron-Phosphate (LFP) cells. Temperature is maintained by a coolant system. The battery capacity is 175 kWh with an operation voltage in the range 450 to 660V.

	  	12V
	    	660V





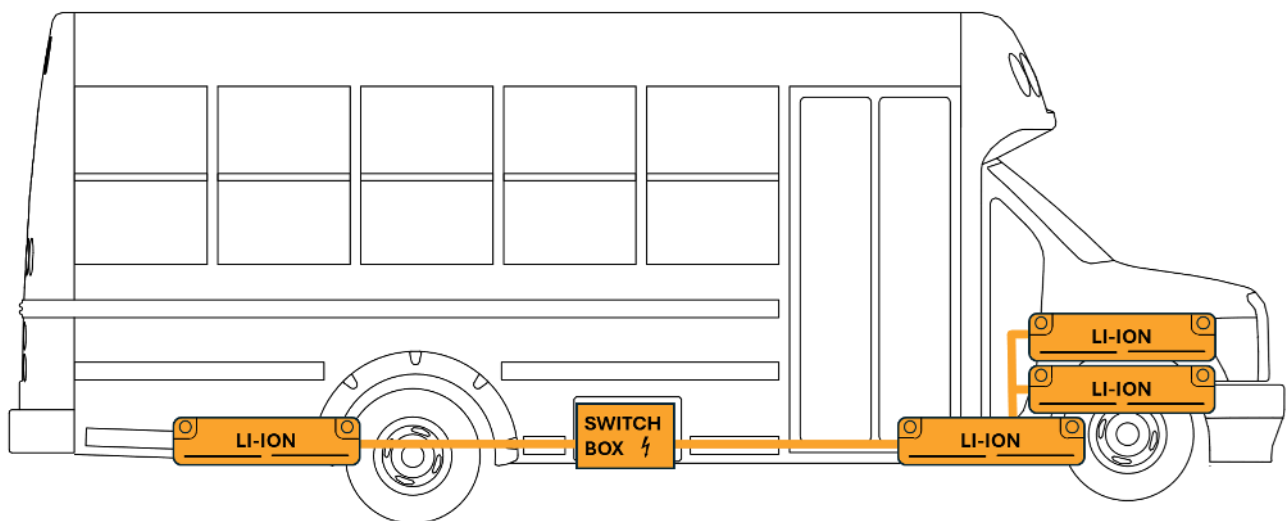
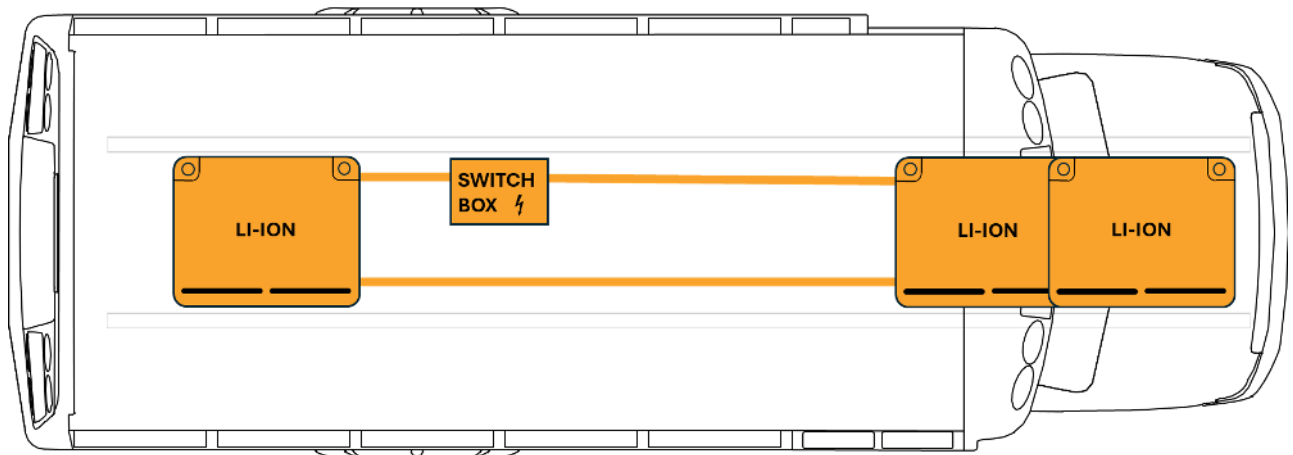
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G5e 140 kWh

The High Voltage (HV) battery is composed of four (4) lithium-ion packs built from Lithium-Iron-Phosphate (LFP) cells. Temperature is maintained by a coolant system. The battery capacity is 140 kWh with an operation voltage in the range 480 to 700V.

	  	12V
	    	700V





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6. In Case of Fire



Park vehicle in a safe area and evacuate immediately when fumes are detected.

Keep passengers and bystanders at least 50 ft (15 m) away from the vehicle until first responders arrive on site.

Make sure nobody stays downwind of (breathes) the fumes coming from the vehicle.

WARNING! ALWAYS wear full PPE including a Self-Contained Breathing Apparatus (SCBA) when fire is involved. Fires in crash-damaged Electric Vehicles (EV) could emit toxic or combustible gases. Small amounts of eye, skin, or lung irritants may be present. If exposed, rinse with large amounts of water for 10-15 minutes. Consider the entire bus as energized at all times. PPE should include: Class 0 electrical insulating gloves, safety goggles, and Electrical Hazard (EH) rated safety shoes or boots, which must be at least appropriate up to 1,000 volts AND a non-conductive object approximately 5 ft (1.5 m) long to safely push someone away from the bus if they accidentally come in contact. Failure to comply can result in serious injury or death.

WARNING! Flames, smoke, arcing, or hot spots like melted plastic may indicate fire or the presence of High Voltage (HV). A thermal camera or Infrared (IR) temperature probe may be useful to identify hot spots. Clear the area around the bus, if possible, and open the bus doors to avoid build-up of gases. If a battery has started burning, it will try to burn to completion. Cool down the batteries with direct and large amounts of water. If possible, remove combustible materials from around the bus to prevent fire growth. Failure to comply can result in serious injury or death.

WARNING! If the batteries catch fire, are exposed to high heat, or are generating heat or gases, use large amounts of water to cool the High Voltage (HV) battery packs. If water is not immediately available, use CO₂, dry chemicals, or another typical fire-extinguishing agent to fight the fire until water is available. Failure to comply can result in serious injury or death.

WARNING! Do NOT use foam to extinguish a fire on electric buses. Failure to comply can result in serious injury or death.

WARNING! Take appropriate measures to protect civilians downwind from the incident. Failure to comply can result in serious injury or death.



7. In Case of Submersion



If the bus is submerged in water, the High Voltage Batteries are designed to resist water intrusion following submersion. There is low risk of electric shock from contacting the bus's body or frame in or out of the water.

Evacuate the bus immediately. Remove key from ignition barrel, if possible.

Do not proceed with any disabling activities until the bus is no longer submerged and drained of water.

Once the bus is no longer submerged, park the bus at least 100 feet (30 m) away from any combustible materials.

It is safe to proceed with disabling procedures when water has been drained, but DO NOT attempt to remove manual service disconnects.

See Disable Direct Hazards /Safety Regulations section of this guide for disabling procedures.

For more information, see [Submerged hybrid/electric vehicles \(NFPA Bulletin\)](#).

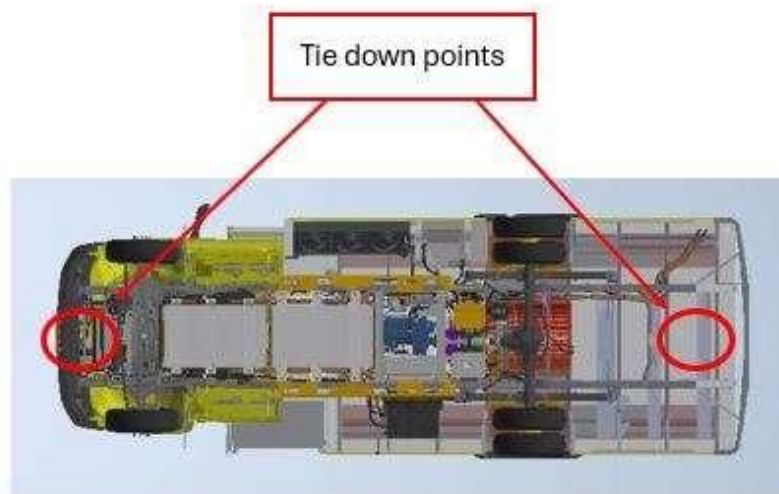


8. Towing / Transportation / Storage

If towing is necessary, the use of a flatbed towing is recommended. If a flatbed towing is not available, rear wheel lift towing is also possible.

Flatbed Towing

If possible, the vehicle can get on the flatbed towing using its own power. If it is not possible, either the transmission should be in neutral position, or the driveshaft should be removed, and winch point should be used. Always mark the location of the driveshaft to the attaching flange before removal so the driveshaft can be correctly indexed upon reinstallation. Replace driveshaft attachment bolts with OEM bolts or add thread locking compound to the original bolts. Once the vehicle is on the flatbed, tie down points in red should be used to secure the vehicle.



Rear Wheel Lift Towing

If a flatbed towing is not available, Rear Wheel Lift Towing is possible. The vehicle should be lifted in the back. Steering should be properly secured. Maximum towing speed is 80 km/h (50 mph), or 55 km/h (35 mph) on a rough road. In the case of a Rear Wheel Lift Tow, the transmission lever's position as no importance and the removal of the driveshaft is not required.

Front Wheel Lift Towing

Front Wheel Lift Towing is **not approved**.



CAUTION! Moving the bus with the rear wheels on the ground and driveshaft not removed, even for a short distance, can cause the drivetrain motor to overheat and lead to irreversible damage of the motor and possibly start a fire.

WARNING! Never attempt to lift the bus by the tow hooks. Tow hooks are designed for horizontal pulling only, NOT FOR LIFTING. Failure to comply can result in serious injury or death.



9. Important Additional Information

Additional information about accident assistance and recovery of the bus with High Voltage (HV) systems can be found at:

- SAE J2990 Hybrid and EV First and Second Responder Recommended Practice
- National Transportation Safety Board (NTSB): www.nts.gov
- National Highway Traffic Safety Administration (NHTSA): www.nhtsa.gov
- U.S. Department of Transportation (USDOT): www.transportation.gov

The High Voltage (HV) LFP battery of the bus requires special handling and disposal. Contact us if disposal is necessary.

Website: www.microbird.com

Phone: (819) 477-8222

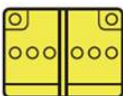
Micro Bird Inc.
3000, Girardin street
Drummondville, QC
Canada J2E 0A1



Emergency Response Guide



10. Explanation of Pictograms Used

				
Electric Propulsion	High Voltage Battery	Electricity Hazard	Arc Flash Hazard	High Voltage Component
				
First Responders Loop	High Voltage Cable	Charge Port	Cooling Component	No Cut Zone
				
Low Voltage Battery	Low Voltage Battery Disconnect	Airbag	Emergency Exit	Flammable
				
Corrosive Substances	Hazardous to Human Health	Acute Toxicity	Explosive	Use Water to Extinguish Fire
				
Thermal Camera				